Efficient Resource Allocation and Power Control for LTE-A D2D Communication With Pure D2D Model

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Introduction

- In general, RBs are considered as exclusive resources from the viewpoint of cellular user equipments (CUEs) [5], which are the UEs talking with the eNB.
- D2D communication offers another way for resource sharing between CUEs and D2D pairs.
- However, not every CUE is suitable to share resources with D2D pairs. In particular, it is difficult for D2D pairs to reuse the RBs allocated to a CUE close to cell edge.
- To deal with this issue, we propose a new sharing paradigm, namely the pure D2D model, to let multiple D2D pairs reuse RBs without the intervention of CUEs.
- By relaxing the assumption that D2D pairs are allowed to reuse only the RBs allocated to CUEs, the pure D2D model facilitates resource sharing among UEs, thereby improving network performance.

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System Model

- We aim at the downlink scenario. Fig. 1 shows the network topology.

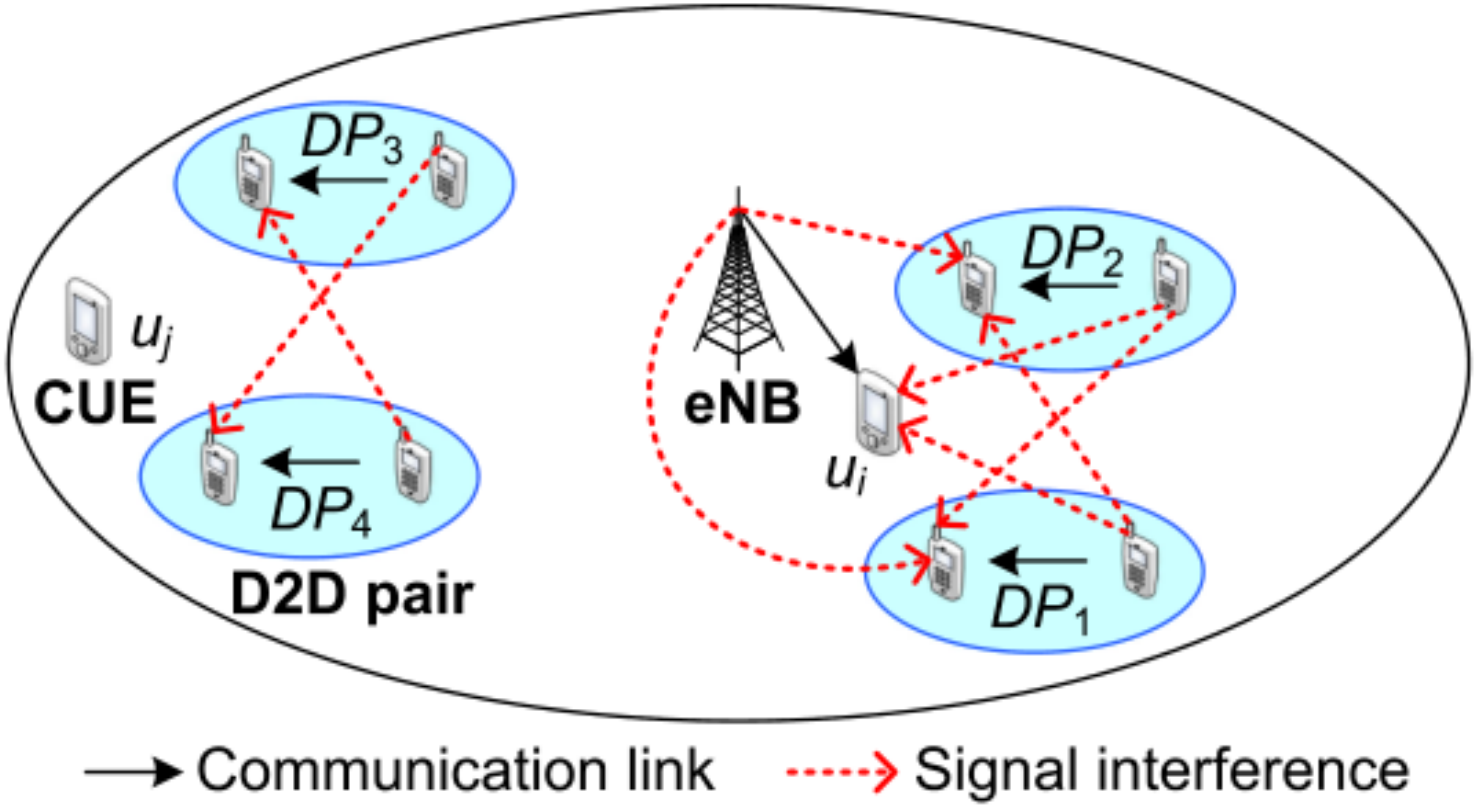


图 1: Example of interference between CUEs and D2D pairs.

System Model

- $\hat{\mathcal{U}}_C$ is the set of CUEs. $\hat{\mathcal{U}}_D$ is the set of receiver of each D2D pair. The union of $\hat{\mathcal{U}}_C$ and $\hat{\mathcal{U}}_D$ is denoted by $\hat{\mathcal{U}}$.
- A CUE is not a D2D sender or receiver. It is obvious that $\hat{\mathcal{U}}_C \cap \hat{\mathcal{U}}_D = \emptyset$.
- The eNB allocates RBs to UEs based on three principles:
  1. Two CUEs cannot use the same RBs.
  2. Multiple D2D pairs may reuse the RBs given to a CUE.
  3. Some RBs could be shared solely by D2D pairs.
- Suppose that a UE u_x receives data from its sender $t(x)$ via RB r_k.
- The strength of $t(x)$' s signal gotten by u_x is $\zeta^k_{t(x),x} = \tilde{g}^k_{t(x),x} \times \tilde{p}^k_{t(x),x}$.
  1. $\tilde{g}^k_{t(x),x}$ is r_k' s channel gain from $t(x)$ to u_x.
  2. $\tilde{p}^k_{t(x),x}$ is $t(x)$' s transmitted power on r_k for sending data to u_x.

System Model

- The amount of interference by $t(x)$ imposed on u_y (denoted by $t(x) \rightarrow u_y$) is calculated as follows:
  1. eNB (sending data to CUE u_x) $\rightarrow$ CUE u_y: $\zeta^k_{t(x),y} = 0$, as r_k cannot be shared by two CUEs.
  2. eNB (sending data to CUE u_x) $\rightarrow$ DUE u_y: $\zeta^k_{t(x),y} = \tilde{g}^k_{eNB,y} \times \tilde{p}^k_{eNB,x}$.
  3. DUE (sending data to DUE u_x) $\rightarrow$ UE u_y: $\zeta^k_{t(x),y} = \tilde{g}^k_{t(x),y} \times \tilde{p}^k_{t(x),x}$, where u_y is a CUE or DUE.

- For a CUE $u_j \in \hat{\mathcal{U}}_C$ its SINR on r_k is estimated by:

$$\tau^k_j = \frac{z^k_j \times \zeta^k_{\mathrm{eNB},j}}{N_0^2 + \sum_{u_i \in \hat{\mathcal{U}}_D} z^k_i \zeta^k_{t(i),j}}, \tag{1}$$

  where N_0 is the power of thermal noise. Z^k_j is an indicator to reveal whether r_k is used by u_j to get data. Here, $Z^k_j = 1$ if r_k is allocated to u_j, $Z^k_j = 0$ otherwise.

- The SINR of a DUE $u_i \in \hat{\mathcal{U}}_D$ on r_k is computed by:

$$\tau^k_i = \frac{z^k_i \times \zeta^k_{t(i),i}}{N_0^2 + \sum_{u_j \in \hat{\mathcal{U}}_C} z^k_j \zeta^k_{\mathrm{eNB},i} + \sum_{u_l \in \hat{\mathcal{U}}_D \setminus \{u_i\}} z^k_l \zeta^k_{t(l),i}}, \tag{2}$$

Problem Formulation

- Given a set $\hat{\mathcal{R}}$ of available RBs, the D2D communication problem is formulated as an optimization problem:

$$\max \sum_{r_k \in \hat{\mathcal{R}}} \sum_{u_i \in \hat{\mathcal{U}}} z_i^k, \tag{3}$$

under the following constraints:

$$\tau_i^k \geq \tau_i^{\min} \text{ if } z_i^k = 1 \quad \forall u_i \in \hat{\mathcal{U}}, \quad \forall r_k \in \hat{\mathcal{R}}, \tag{4}$$

$$z_i^k \in \{0, 1\} \quad \forall u_i \in \hat{\mathcal{U}}, \quad \forall r_k \in \hat{\mathcal{R}}, \tag{5}$$

$$\tilde{p}_{t(i),i}^k \in [p_i^{\min}, p_i^{\max}] \quad \forall u_i \in \hat{\mathcal{U}}, \quad \forall r_k \in \hat{\mathcal{R}}, \tag{6}$$

$$\sum_{u_i \in \hat{\mathcal{U}}_C} z_i^k \leq 1 \quad \forall r_k \in \hat{\mathcal{R}}. \tag{7}$$

- The objective in Eq. 3 is to maximize the number of communication links supported by RBs in $\hat{\mathcal{R}}$.

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Color-Based Grouping Algorithm

- DRAPC adopts a graph $\mathcal{H} = (\mathcal{V}, \mathcal{E})$ to reveal interference among UEs, where the vertex set $\mathcal{V}$ includes every UE in $\hat{\mathcal{U}}$.
- If a UE u_i will be interfered by the sender of another UE u_j, there exists an edge (u_i, u_j) in $\mathcal{E}$. In this case, we say that u_i is a neighbor of u_j.
- This is similar to the vertex coloring problem [37], where two vertices linking by the same edge cannot have the same color.
- When a group of vertices are painted with the same color, the corresponding UEs are interference-free and allowed to use the same RBs.
- DRAPC seeks to exchange and add members in each group to help more UEs share resources, and adjusts transmitted power to raise their SINRs.

Color-Based Grouping Algorithm

- This algorithm divides UEs into groups by vertex coloring such that all members in each group can share RBs without interference.
- Algorithm 1 gives its pseudocode that contains three parts.
- Part 1 (lines 1–4) builds the graph, where $\hat{\mathcal{N}}_i$ is the set of neighbors of a UE u_i in $\hat{\mathcal{U}}$. ζ_{th} is a threshold.
- Part 2 includes lines 5 to 12, which adopts a greedy strategy to group UEs. Each RB r_k in $\hat{\mathcal{R}}$ is viewed as one color, and we use $\hat{\mathcal{G}}_k$ to record the set of UEs that share r_k.
- All UEs of $\hat{\mathcal{U}}$ are sorted by the number of neighbors (i.e., $|\hat{\mathcal{N}}_i|$) in a descending order. For each UE u_i in $\hat{\mathcal{C}}$, line 7 sorts each RB by its signal strength $\zeta^k_{t(i)}$ decreasingly.
- Prt 3 calls the power control algorithm in Algorithm 3 to estimate transmitted power.

Color-Based Grouping Algorithm

Algorithm 1: Color-based Grouping Algorithm.

foreach $u_i \in \hat{\mathcal{U}}$ **do**
  foreach $u_j \in \hat{\mathcal{U}}$ **do**
    if $u_i \neq u_j$ *and* $\max_{r_k \in \hat{\mathcal{R}}} \zeta^k_{t(i),j} > \zeta_{\text{th}}$ **then**
      $\hat{\mathcal{N}}_i \leftarrow \hat{\mathcal{N}}_i \cup \{u_j\}$ and $\hat{\mathcal{N}}_j \leftarrow \hat{\mathcal{N}}_j \cup \{u_i\}$;
$\hat{\mathcal{C}} \leftarrow \texttt{SORT}(\hat{\mathcal{U}}, |\hat{\mathcal{N}}_i|)$;
foreach $u_i \in \hat{\mathcal{C}}$ **do**
  $\hat{\mathcal{R}}_i \leftarrow \texttt{SORT}(\hat{\mathcal{R}}, \zeta^k_{t(i),i})$;
  foreach $r_k \in \hat{\mathcal{R}}_i$ **do**
    if $\hat{\mathcal{G}}_k \cap \hat{\mathcal{N}}_i = \emptyset$ **then**
      if $u_i \notin \hat{\mathcal{U}}_C$ *or* $\hat{\mathcal{G}}_k \cap \hat{\mathcal{U}}_C = \emptyset$ **then**
        $\hat{\mathcal{G}}_k \leftarrow \hat{\mathcal{G}}_k \cup \{u_i\}$ and $\hat{\mathcal{C}} \leftarrow \hat{\mathcal{C}} \setminus \{u_i\}$;
        break;
foreach $\hat{\mathcal{G}}_k \subseteq \hat{\mathcal{U}}$ **do**
  Adjust transmitted power in $\hat{\mathcal{G}}_k$ by Algo. 3;
  while $\min_{u_i \in \hat{\mathcal{G}}_k} \tau^k_i < \tau^{\min}_i$ **do**
    $u_i \leftarrow \arg\min_{u_i \in \hat{\mathcal{G}}_k} \tau^k_i$;
    $\hat{\mathcal{G}}_k \leftarrow \hat{\mathcal{G}}_k \setminus \{u_i\}$ and $\hat{\mathcal{C}} \leftarrow \hat{\mathcal{C}} \cup \{u_i\}$;

Group Reforming Algorithm

- Algorithm 1 groups UEs for RB allocation by their relationship of interference, but some UEs may be still left unallocated in the set $\hat{\mathcal{C}}$.
- The group reforming algorithm aims to improve each group by exchanging its members with UEs in $\hat{\mathcal{C}}$ that have better conditions.
- $\hat{\mathcal{G}}_{tmp}$ is a temporary group. Line 10 uses Algorithm 3 to compute transmitted power for UEs in $\hat{\mathcal{G}}_{tmp}$. If both conditions in line 11 are satisfied, a better group $\hat{\mathcal{G}}_{tmp}$ is found for $\hat{\mathcal{G}}_k$.
- After the above adjustment, the member adding algorithm in Algorithm 4 finds extra UEs from $\hat{\mathcal{C}}$ and adds them to group $\hat{\mathcal{G}}_k$.
- The throughput raising algorithm in Algorithm 6 increases transmitted power to improve throughput of UEs.

Group Reforming Algorithm

Algorithm 2: Group Reforming Algorithm.

foreach $\hat{\mathcal{G}}_k \subseteq \hat{\mathcal{U}}$ **do**
  foreach $u_i \in \hat{\mathcal{C}}$ **do**
    if $u_i \in \hat{\mathcal{U}}_C$ *and* $\hat{\mathcal{G}}_k \cap \hat{\mathcal{U}}_C \neq \emptyset$ **then**
      $u_j \leftarrow \hat{\mathcal{G}}_k \cap \hat{\mathcal{U}}_C$;
      if $\tilde{p}^k_{\mathrm{eNB},i} < \tilde{p}^k_{\mathrm{eNB},j}$ **then**
        $\hat{\mathcal{G}}_{\mathrm{tmp}} \leftarrow (\hat{\mathcal{G}}_k \setminus \{u_j\}) \cup \{u_i\}$;
    else if $u_i \in \hat{\mathcal{U}}_D$ **then**
      $u_j \leftarrow \arg\max_{u_j \in \hat{\mathcal{G}}_k} \tilde{p}^k_{t(j),j}$;
      $\hat{\mathcal{G}}_{\mathrm{tmp}} \leftarrow (\hat{\mathcal{G}}_k \setminus \{u_j\}) \cup \{u_i\}$;
    Adjust transmitted power in $\hat{\mathcal{G}}_{\mathrm{tmp}}$ by Algo. 3;
    if $\min_{u_x \in \hat{\mathcal{G}}_{\mathrm{tmp}}} \tau^k_x > \tau^{\min}_x$ *and*
     $\max_{u_x \in \hat{\mathcal{G}}_{\mathrm{tmp}}} \tilde{p}^k_{t(x),x} < \max_{u_y \in \hat{\mathcal{G}}_k} \tilde{p}^k_{t(y),y}$ **then**
      $\hat{\mathcal{G}}_k \leftarrow \hat{\mathcal{G}}_{\mathrm{tmp}}$ and $\hat{\mathcal{C}} \leftarrow (\hat{\mathcal{C}} \setminus \{u_i\}) \cup \{u_j\}$;
  Add extra members to $\hat{\mathcal{G}}_k$ by Algo. 4;
  Raise throughput of UEs in $\hat{\mathcal{G}}_k$ by Algo. 6;

Power Control Algorithm

- For each u_i, line 2 sets the default transmitted power of its sender by

$$\tau_i^k = \frac{\tilde{g}_{t(i),i}^k \times \tilde{p}_{t(i),i}^k}{N_0} \geq \tau_i^{\min} \Rightarrow \tilde{p}_{t(i),i}^k \geq \tau_i^{\min} \times N_0 / \tilde{g}_{t(i),i}^k.$$

- Then, lines 3–6 do the boundary check for the constraint in Eq. 6.
- In lines 8–13, we adopt a while-loop to adjust power $\tilde{p}_{t(i),i}^k$, by a shrinking variable p_{adj}.
- We remark that the objective of the power control algorithm in Algorithm 3 is to find the smallest transmitted power for a node $t(i)$ to let the SINR of its receiver u_i overtake τ_{min}.

Power Control Algorithm

Algorithm 3: Power Control Algorithm.

foreach $u_i \in \hat{\mathcal{G}}_k$ **do**
  $\tilde{p}^k_{t(i),i} \leftarrow \tau_i^{\min} \times N_0 / \tilde{g}^k_{t(i),i}$;
  if $\tilde{p}^k_{t(i),i} > p_i^{\max}$ **then**
    $\tilde{p}^k_{t(i),i} \leftarrow p_i^{\max}$;
  else if $\tilde{p}^k_{t(i),i} < p_i^{\min}$ **then**
    $\tilde{p}^k_{t(i),i} \leftarrow p_i^{\min}$;
  $p_{\text{adj}} \leftarrow p_i^{\max} / \sigma$;
  while $p_{\text{adj}} \geq p_i^{\max} / \sigma^{\alpha}$ **do**
    if $\tau_i^k < \tau_i^{\min}$ *and* $\tilde{p}^k_{t(i),i} + p_{\text{adj}} \leq p_i^{\max}$ **then**
      $\tilde{p}^k_{t(i),i} \leftarrow \tilde{p}^k_{t(i),i} + p_{\text{adj}}$;
    else if $\tau_i^k > \tau_i^{\min} \times (1 + (p_{\text{adj}} / \tilde{p}^{\min}_{t(i),i}))$ *and* $\tilde{p}^k_{t(i),i} - p_{\text{adj}} \geq p_i^{\min}$ **then**
      $\tilde{p}^k_{t(i),i} \leftarrow \tilde{p}^k_{t(i),i} - p_{\text{adj}}$;
    $p_{\text{adj}} \leftarrow p_{\text{adj}} / \sigma$;

Member Adding Algorithm

- This algorithm picks extra members for a group $\hat{\mathcal{G}}_k$ from the set $\hat{\mathcal{C}}$, which contains the UEs that have not yet been assigned to any group.
- Algorithm 4 presents its pseudocode, where we copy $\hat{\mathcal{C}}$ to a temporary set $\hat{\mathcal{C}}_{tmp}$.
- After eliminating non-candidates from $\hat{\mathcal{C}}_{tmp}$, we adopt the branch-and-bound (BnB) method [39] to search new members for $\hat{\mathcal{G}}_k$ (the updated set is denoted by $\hat{\mathcal{G}}_{new}$), and remove these UEs from $\hat{\mathcal{C}}$.
- The BnB method builds a state space tree, which is a binary tree whose nodes indicate whether to add each UE in $\hat{\mathcal{C}}_{tmp}$ to $\hat{\mathcal{G}}_{new}$.
- BnB uses the breadth-first search to check each node in the state space tree.
- Since the tree size could be very large, a bound function is adopted to do the check. If a node cannot be passed by the bound function, it means that including this node will make the solution worse.

Member Adding Algorithm

- Fig. 2 shows an example.

Algorithm 4: Member Adding Algorithm.

$\hat{\mathcal{C}}_{\text{tmp}} \leftarrow \hat{\mathcal{C}}$;
if $\hat{\mathcal{G}}_k \cap \hat{\mathcal{U}}_C \neq \emptyset$ **then**
  $\hat{\mathcal{C}}_{\text{tmp}} \leftarrow \hat{\mathcal{C}}_{\text{tmp}} \setminus \hat{\mathcal{U}}_C$;
foreach $u_i \in \hat{\mathcal{C}}_{\text{tmp}}$ **do**
  Adjust transmitted power in $\hat{\mathcal{G}}_k \cup \{u_i\}$ by Algo. 3;
  if $\min_{u_j \in \hat{\mathcal{G}}_k \cup \{u_i\}} \tau_j^k < \tau_j^{\min}$ **then**
    $\hat{\mathcal{C}}_{\text{tmp}} \leftarrow \hat{\mathcal{C}}_{\text{tmp}} \setminus \{u_i\}$;
Use BnB to select new members from $\hat{\mathcal{C}}_{\text{tmp}}$ and add them to $\hat{\mathcal{G}}_k$; the result is stored in $\hat{\mathcal{G}}_{\text{new}}$;
$\hat{\mathcal{G}}_k \leftarrow \hat{\mathcal{G}}_{\text{new}}$ and $\hat{\mathcal{C}} \leftarrow \hat{\mathcal{C}} \setminus \hat{\mathcal{G}}_{\text{new}}$;

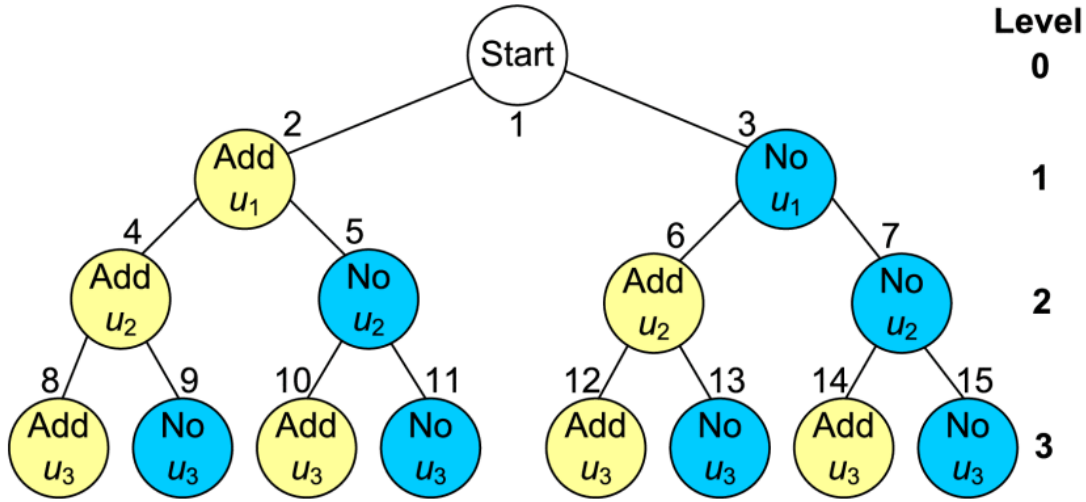


图 2: Example of a state space tree.

Member Adding Algorithm

- Algorithm 5 gives the pseudocode of our bound function, where p_{low} and n_{max} record the maximum transmitted power and the number of UEs in the best solution $\hat{\mathcal{G}}_{new}$ that we have found, respectively.

Algorithm 5: Bound Function for BnB.

Data: $p_{\text{low}} \leftarrow \infty$, $n_{\max} \leftarrow |\hat{\mathcal{G}}_k|$, $\hat{\mathcal{G}}_{\text{new}} \leftarrow \hat{\mathcal{G}}_k$

if *the visiting node is a left child in level* i **then**
  $\hat{\mathcal{G}}_{\text{new}} \leftarrow \hat{\mathcal{G}}_{\text{new}} \cup \{u_i\}$;
Adjust transmitted power in $\hat{\mathcal{G}}_{\text{new}}$ by Algo. 3;
if $\min_{u_j \in \hat{\mathcal{G}}_{\text{new}}} \tau_j^k < \tau_j^{\min}$ **then**
  return false;
if $|\hat{\mathcal{G}}_{\text{new}}| > n_{\max}$ **then**
  $n_{\max} \leftarrow |\hat{\mathcal{G}}_{\text{new}}|$ and $p_{\text{low}} \leftarrow \max_{u_j \in \hat{\mathcal{G}}_{\text{new}}} \tilde{p}^k_{t(j),j}$;
  return true;
else if $|\hat{\mathcal{G}}_{\text{new}}| = n_{\max}$ *and* $\max_{u_j \in \hat{\mathcal{G}}_{\text{new}}} \tilde{p}^k_{t(j),j} < p_{\text{low}}$ **then**
  $p_{\text{low}} \leftarrow \max_{u_j \in \hat{\mathcal{G}}_{\text{new}}} \tilde{p}^k_{t(j),j}$;
  return true;
return false;

Throughput Raising Algorithm

- After finding all members of a group $\hat{\mathcal{G}}_k$, the last algorithm seeks to raise transmitted power for some UEs to improve the overall throughput in $\hat{\mathcal{G}}_k$.
- Algorithm 6 gives its pseudocode, where $\hat{\mathcal{C}}_{tmp}$ is a subset of UEs in $\hat{\mathcal{G}}_k$ whose senders are capable of raising power.
- We compute the amount of throughput gain in $\hat{\mathcal{G}}_k$ (denoted by ζ_i^{inc}) when $t(i)$ uses the new power $\tilde{p}_i^{new}$ by line 8.
- Let ζ_j^{new} and ζ_j denote the amount of throughput of a UE u_j in $\hat{\mathcal{G}}_k$ when $t(i)$' s power is $\tilde{p}_i^{new}$ and $\tilde{p}_{(t)i,i}^{k}$, respectively.
- If ζ_i^{inc} is not positive, which implies that there is no throughput gain by using the new power $\tilde{p}_i^{new}$, u_i is removed from $\hat{\mathcal{C}}_{tmp}$.

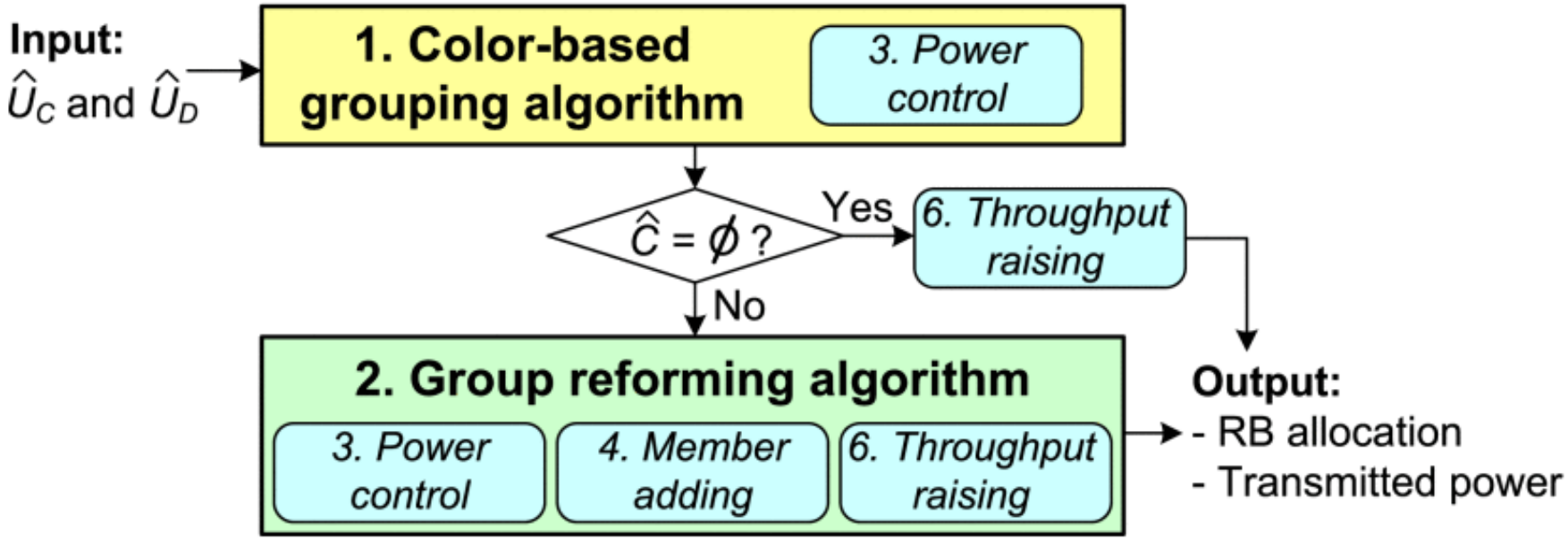
Power Control Algorithm

Algorithm 6: Throughput Raising Algorithm.

$\hat{\mathcal{G}}_{\text{tmp}} \leftarrow \hat{\mathcal{G}}_k$;
while *true* **do**
  foreach $u_i \in \hat{\mathcal{G}}_{\text{tmp}}$ **do**
    $\tilde{p}_i^{\text{new}} \leftarrow \tilde{p}_{t(i),i}^{k} + \delta$;
    if $\tilde{p}_i^{\text{new}} > p_i^{\max}$ *or* $\min_{u_j \in \hat{\mathcal{G}}_k} \tau_j^k < \tau_j^{\min}$ **then**
      $\hat{\mathcal{G}}_{\text{tmp}} \leftarrow \hat{\mathcal{G}}_{\text{tmp}} \setminus \{u_i\}$;
    else
      $\xi_i^{\text{inc}} \leftarrow \sum_{u_j \in \hat{\mathcal{G}}_k} (\xi_j^{\text{new}} - \xi_j)$;
      if $\xi_i^{\text{inc}} \leq 0$ **then**
        $\hat{\mathcal{G}}_{\text{tmp}} \leftarrow \hat{\mathcal{G}}_{\text{tmp}} \setminus \{u_i\}$;
  if $\hat{\mathcal{G}}_{\text{tmp}} = \emptyset$ **then**
    break;
  $u_i \leftarrow \arg\max_{u_i \in \hat{\mathcal{G}}_{\text{tmp}}} \xi_i^{\text{inc}}$;
  $\tilde{p}_{t(i),i}^{k} \leftarrow \tilde{p}_i^{\text{new}}$;

The DRAPC Framework

- The DRAPC framework is shown in Fig. 3.



- We use Algorithm 1 to group UEs.

- If the candidate set $\hat{\mathcal{C}}$ becomes empty, which means that every UE in $\hat{\mathcal{U}}$ has been allocated with RBs by Algorithm 1, Algorithm 6 is called to improve throughput by raising transmitted power.

The DRAPC Framework

- When $\hat{\mathcal{C}}$ is not empty after the execution of Algorithm 1, the Algorithm 2 then reforms each group by exchanging its members and the unassigned UEs in $\hat{\mathcal{C}}$ to improve signal quality in the group.
- Algorithm 3 adjusts the transmission power to meet the SINR requirements of the terminal.
- Algorithm 4 checks if extra members (from $\hat{\mathcal{C}}$) can be included in each group to support more links.
- Though Algorithm 4 uses the BnB method (i.e., Algorithm 5) to search extra members.
- Afterwards, Algorithm 6 is used to maximize network throughput.

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Simulation

eNB's parameters:	
channel bandwidth	5 MHz (supports 25 RBs per TTI)
cell range	500 meters
transmitted power	min: -40 dBm, max: 46 dBm
frame structure	frequency division duplexing (FDD)
modulation	QPSK, 16QAM, 64QAM
UE's parameters:	
number of CUEs	10 and 25
number of D2D pairs	small-scale: 5–25, large-scale: 25–75
transmitted power	min: -40 dBm, max: 23 dBm
SINR threshold $\tau_i^{\min}$	10 dB and 15 dB
channel's parameters:	
propagation loss	urban macro-cell model
path loss	eNB to CUE: $128.1 + 37.6 \log L$
	D2D pair: $148 + 40 \log L$
slow fading	shadowing fading:
	log-normal distribution (mean: 0 dB, SD: 8 dB)
fast fading	the Rayleigh fading model
thermal noise N_0	-174dBm/Hz

- The effect of path loss depends on the distance L between two nodes. For a D2D pair, L ranges from 1 to 20 meters.

- We compared with three methods, the Gale-Shapley (GS) method the graph coloring based resource sharing (GCRS) method and color-based grouping and throughput raising (CBG-TR) method.

Simulation

- We first observe the gap between the optimal solution and each method, where the optimal solution is derived by solving the MINP problem defined in Section IV-C.

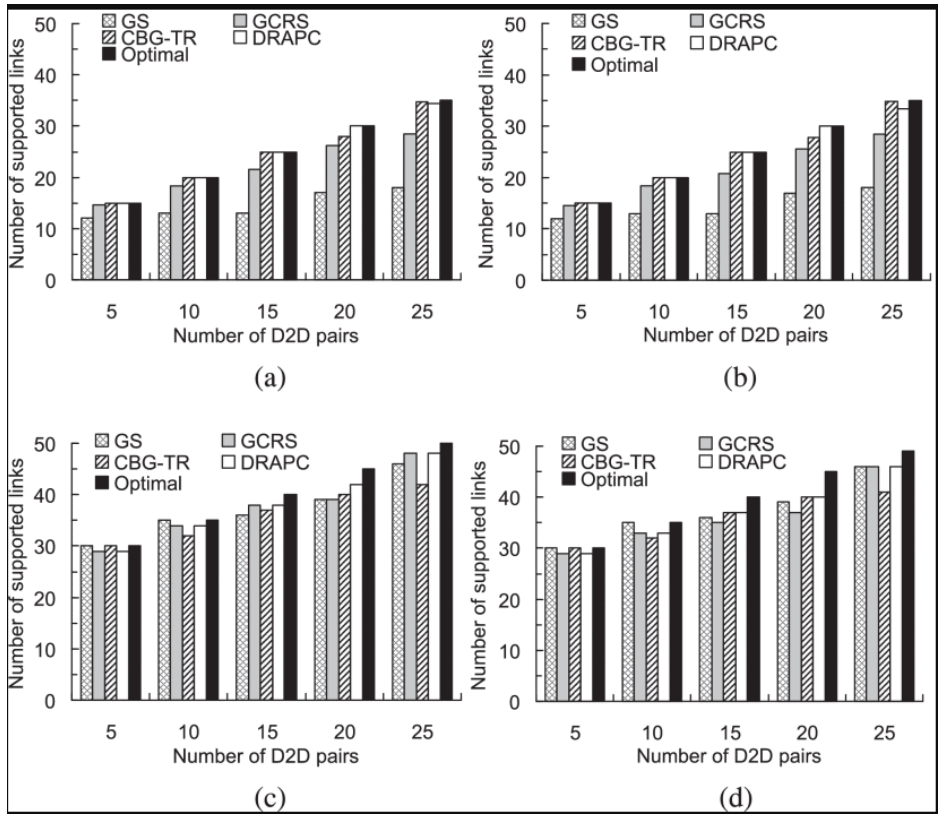
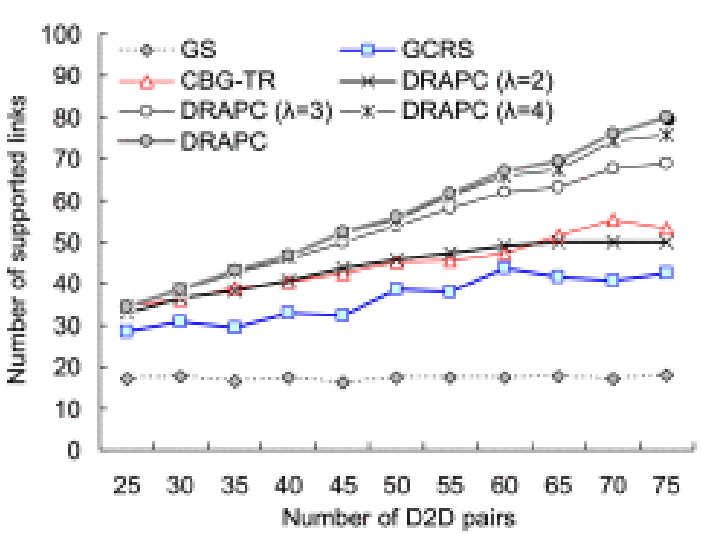


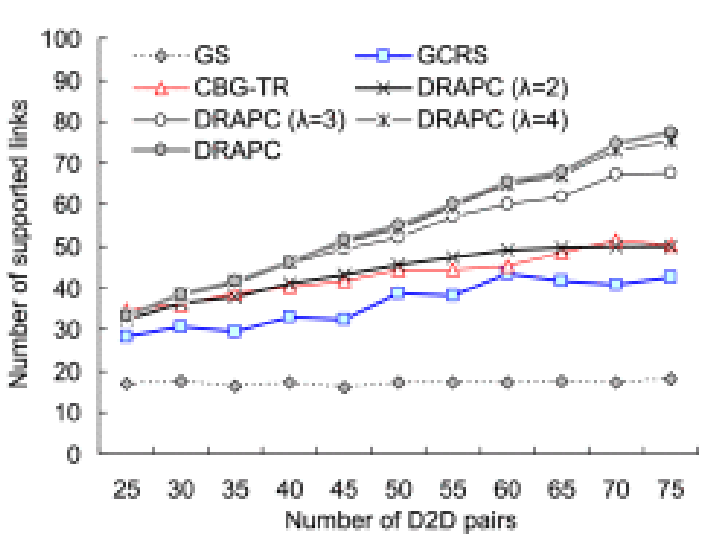
图 3: Comparison on the number of supported links between each method and the optimal solution in a small-scale network. (a) 10 CUEs, τ_i^{min} : 10 dB. (b) 10 CUEs, τ_i^{min} : 15 dB. (c) 25 CUEs, mini : 10 dB. (d) 25 CUEs, τ_i^{min} : 15 dB.

Simulation

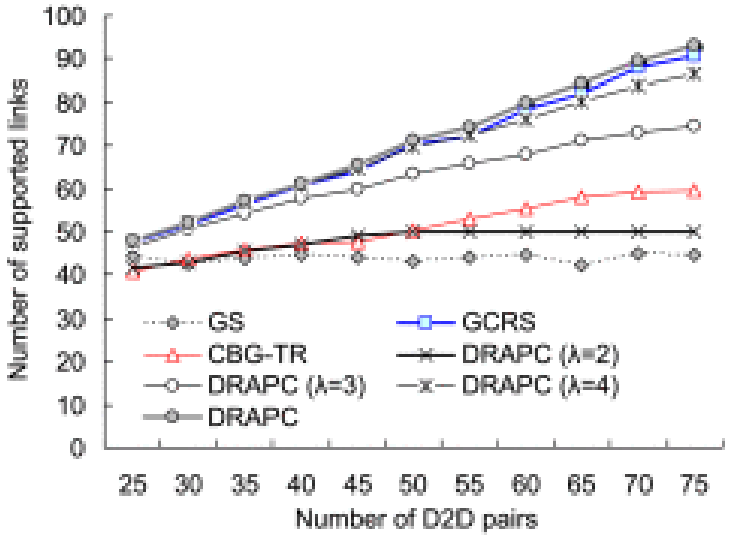
- Fig. 6(a)–(c) present the number of links supported by each method.



(a)



(b)

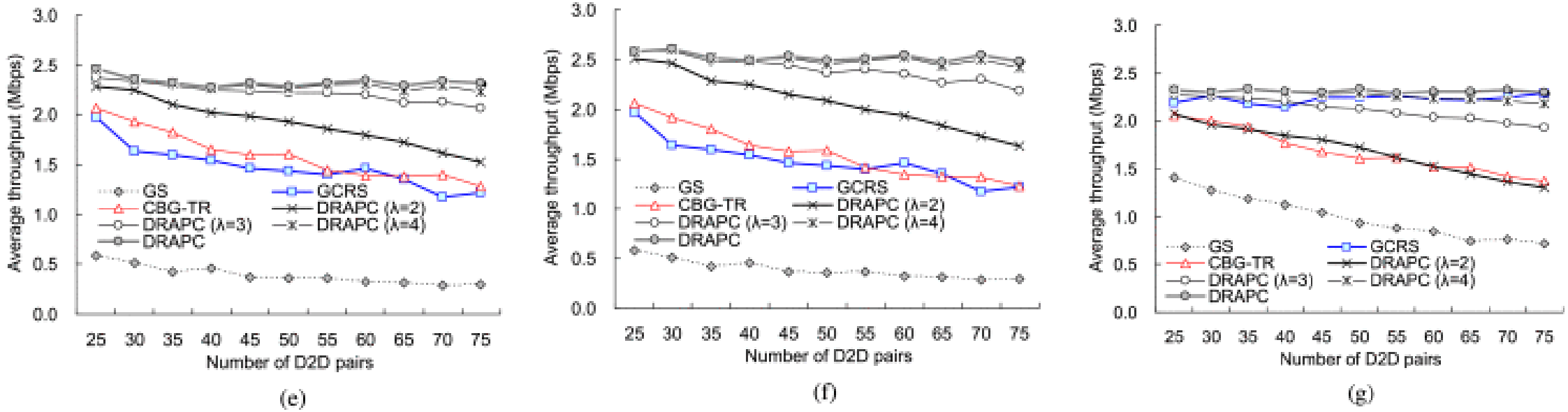


(c)

- (a) Supported links (10 CUEs, τ_i^{min} : 10 dB); (b) Supported links (10 CUEs, τ_i^{min} : 15 dB); (c) Supported links (25 CUEs, τ_i^{min} : 10 dB).

Simulation

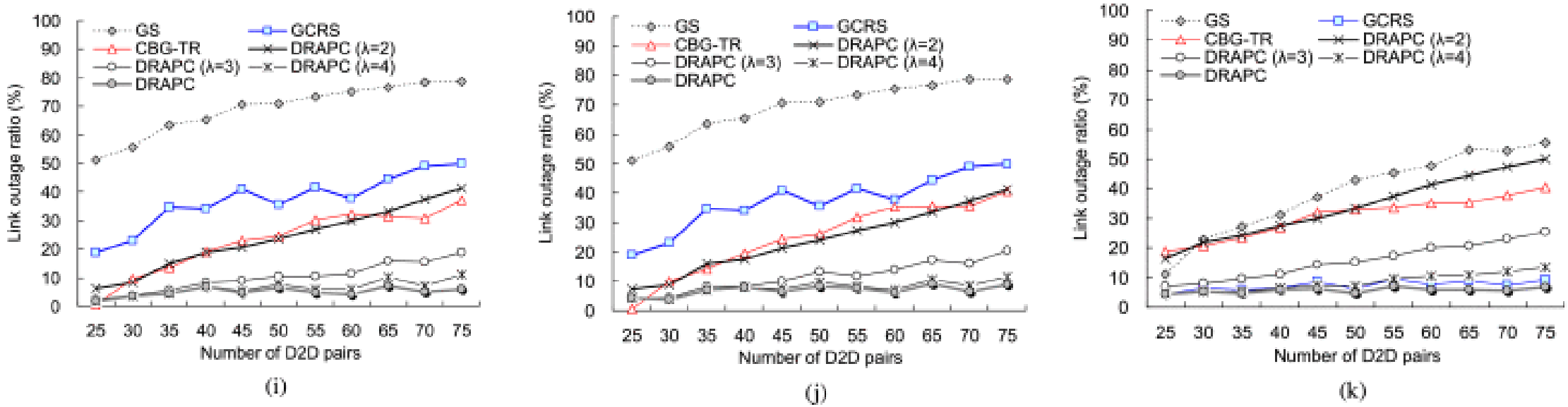
- Fig. 6(e)–(g) measure the average throughput of each link.



- (e) Throughput (10 CUEs, τ_i^{min} : 10 dB); (f) Throughput (10 CUEs, τ_i^{min} : 15 dB); (g) Throughput (25 CUEs, τ_i^{min} : 10 dB).

Simulation

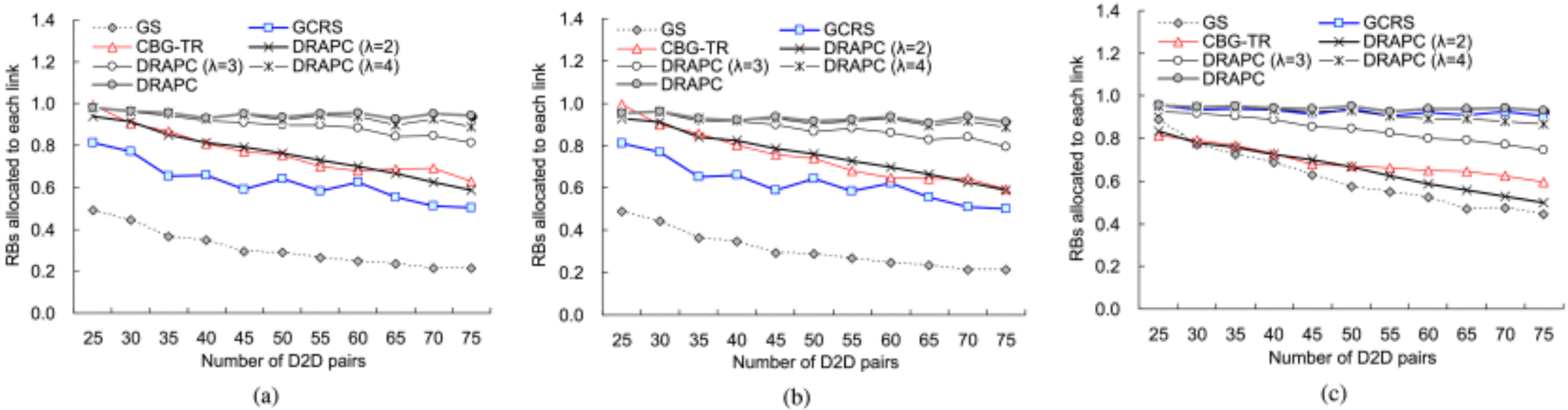
- Fig. 6(i)–(k) evaluate the link outage ratio, where a link is considered as outage if it gets no RB to transmit data.



- (i) Link outage (10 CUEs, τ_i^{min} : 10 dB); (j) Link outage (10 CUEs, τ_i^{min} : 15 dB); (k) Link outage (25 CUEs, τ_i^{min} : 10 dB).

Simulation

- Fig. 7(a)–(c) give the average number of RBs allocated to each link.



- (a) RB allocation (10 CUEs, τ_i^{min} : 10 dB); (b) RB allocation (10 CUEs, τ_i^{min} : 15 dB); (c) RB allocation (25 CUEs, τ_i^{min} : 10 dB).

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Conclusion

- This paper proposes a pure D2D model to provide flexibility for resource allocation, which allows some RBs to be shared solely by DUEs.
- Based on the new model, an efficient DRAPC framework is proposed with the objective of maximizing the number of supported links.
- It groups UEs by vertex coloring, reforms each group through BnB, and raises transmitted power to better SINRs.
- Simulation results show that DRAPC well exploits the pure D2D model to increase throughput, reduce link outage, raise resource utilization, and improve fairness, as compared with GS and graph-coloring based methods.
- Furthermore, we can save the computational cost of DRAPC while achieving a similar performance by setting $\lambda = 4$.

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W. -K. Lai, Y. -C. Wang, H. -C. Lin and J. -W. Li, "Efficient Resource Allocation and Power Control for LTE-A D2D Communication With Pure D2D Model," in IEEE Transactions on Vehicular Technology, vol. 69, no. 3, pp. 3202-3216, March 2020, doi: 10.1109/TVT.2020.2964286.